

Instructions to candidates: Question no. 1 is compulsory; solve any five from Q.2 to 7

If any data is missing is a part of question. Assume suitable data giving reason

Q1. Explain modelling of power system components.

[10marks]

Q2.

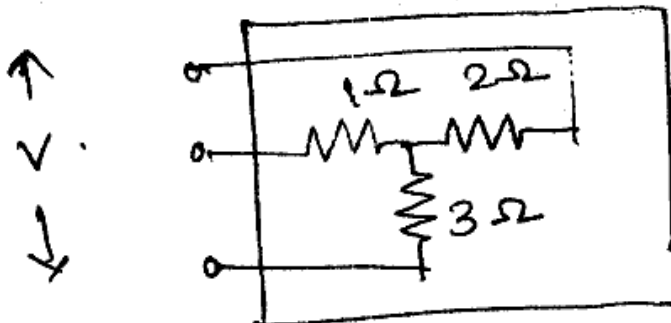
[6+6=12marks]

- Explain equal area criterion considering fault at sending end of system.
- Deduce flow chart for load flow solution using decoupled load flow method.

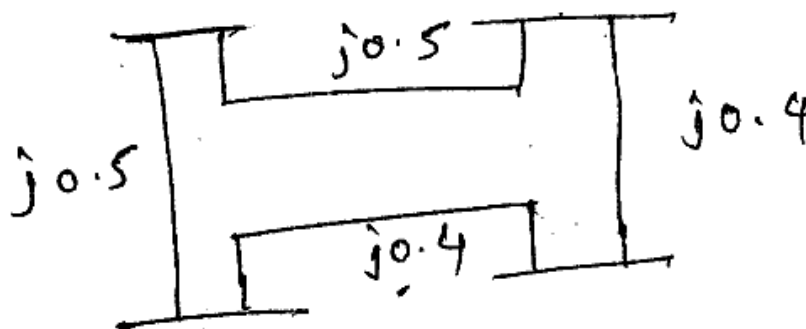
Q3.

[4+8=12marks]

- Formulate Z_{bus} matrix for a two port network.



- Formulate Z_{bus} matrix for given power system.



Q4.

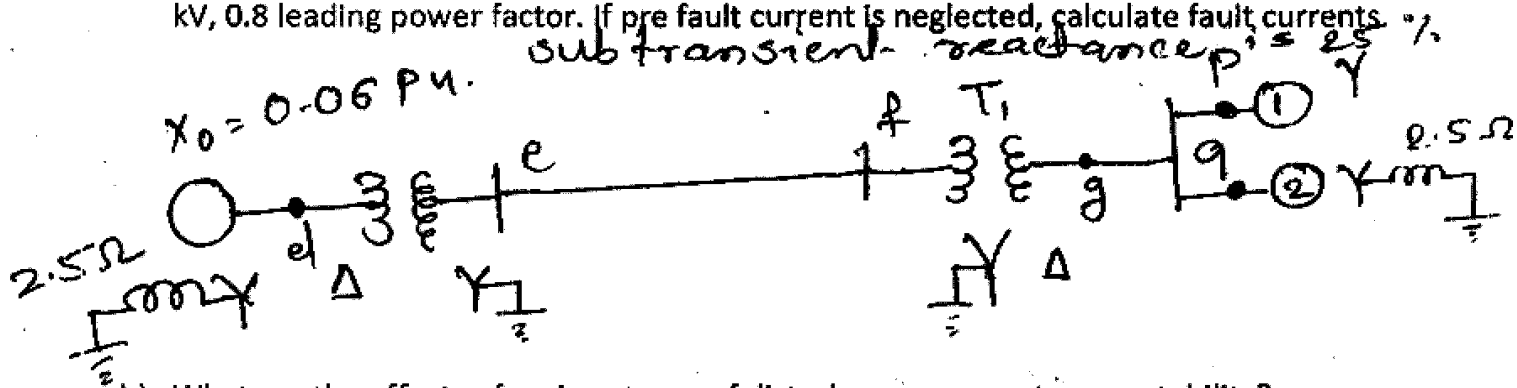
[6+6=12marks]

- Draw functional block diagram of a typical excitation control system for large synchronous generator and explain.
- Enlist different methods of excitation control. With neat schematic explain operation of DC excitation system.

Q5.

[6+6=12marks]

- The one line diagram is shown in fig below. On a base of 25 MVA AND 11KV in generator circuit, draw the positive, negative and zero sequence networks of the system. Before occurrence of a solid LG at bus g, the motors are loaded to draw 15 and 7.5 MW at 10 kV, 0.8 leading power factor. If pre fault current is neglected, calculate fault currents.



- What are the effects of various types of disturbance parameters on stability?

Q6.

[6+6=12marks]

- Draw and explain functional block diagram of power generation and control system shows location of steam turbine and governor.
- From fundamentals explain steam turbine modeling.

Q7.

[4+4+4=12marks]

Write short note on

- Washout block
- Limiter
- Phase compensator